Ayush Madhav Kumar

Ann Arbor, Michigan

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Education

University of Michigan, College of Engineering — **Aug 2025 – December 2028**
Bachelor of Science in Computer Science; Minor in Mathematics — *Ann Arbor, Michigan*

Experience

Jaseci Labs — **May 2026 – Present**
Software Engineering Intern — *Ann Arbor, Michigan*

– Engineering a domain-specialized coding agent for the Jac programming language, fine tuning Gemma 4 26B with Unsloth and 4-bit QLoRA on an object-spatial superset of Python where general-purpose models lack usable priors
– Architecting a multi-recipe synthetic data pipeline with Python-to-Jac parallel translation, adversarial DPO negatives, evol-instruct, and self-distillation, targeting 300k+ examples verified behind a compiler and unit-test hard-gate
– Designing a four-stage training curriculum spanning core SFT, debugging and reasoning, DPO alignment, and multi-turn agentic data to prevent Python-pattern fallback and lock in idiomatic walker, node, and edge constructs

Pathways for Open Source Ecosystems (POSE) Research — **May 2026 – Present**
Professor Misc. (Temp.) / Student Researcher — *Ann Arbor, Michigan*

– Part of NSF Phase I POSE research ($300K) to grow an open-source ecosystem around the data spatial Jac language
– Enhancing the project's digital infrastructure, documentation site, CI, and contributor onboarding tooling to lower the barrier to entry for new developers and grow an active, distributed Jac contributor community
– Authoring tutorials, code samples, and educational materials, and analyzing developer feedback and usage telemetry to refine onboarding workflows and inform governance models for long-term ecosystem sustainability

Cactus (YC S25) — **January 2026 – Present**
Core Contributor — *Ann Arbor, Michigan*

– Contributing to an open source mobile AI inference engine optimized for smartphones and low power devices
– Implementing ARM specific SIMD kernels for matrix multiplication and attention mechanisms with KV cache quantization
– Developing zero copy computation graphs for efficient on device transformer inference with sub 50ms time to first token
– Building OpenAI compatible FFI APIs for cross platform SDK integration across Flutter, React Native, and Kotlin

GIIT Solutions — **June 2024 – July 2024, April 2025**
Associate Software Intern — *Remote*

– Maintained CI/CD pipelines on AWS, automating deployments across 12+ microservices and improving reliability by 28%
– Developed Python scripts to streamline backend processes, reducing manual overhead by 40% and saving 15+ hours weekly
– Collaborated with senior devs to optimize cloud infrastructure, achieving 35% cost reduction through resource management
– Deployed cloud native solutions with AWS CI/CD tools, improving system scalability and achieving 99.7% uptime

Projects/Research

NASA SUITS Challenge | ***AI Lead; Python, PyTorch, A*/JPS, Unity***

– Built autonomous lunar-EVA navigation with A*/Jump Point Search planners over a 3-layer occupancy grid, 17-ray LiDAR fusion, PD heading control with curvature lookahead, and RSSI trilateration via inverse log-distance modeling
– Designed PyTorch obstacle detection models, a temporal ResidualBeamCNN and a BEV cross-attention transformer with learnable query embeddings, served in real time behind a hot-swappable multi-backend inference framework
– Engineered the voice pipeline, streaming 16 kHz PCM over WebSocket to a faster-whisper and a neural intent classifier

Machine Learning Model Serving Microservice | ***Python, Flask, RabbitMQ, MongoDB, Docker***

– Developed a backend microservice to host ML models through Flask REST APIs, for model loading, inference, health checks
– Handled inference requests using RabbitMQ for scalable task queuing, enabling reliable processing of high-volume workloads
– Persisted model outputs and user queries in MongoDB for analytics and logging, and containerized the system using Docker

Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScript, Java, C, C++, Go, Rust, Ruby, SQL, Jac, HTML, CSS
- **AI and ML:** PyTorch, TensorFlow, JAX, HuggingFace Transformers, Unsloth, LoRA/QLoRA fine tuning, RLHF/DPO, ONNX Runtime, vLLM, Scikit learn, OpenAI APIs, LLMs, RAG, Mixture of Experts, NLP, Computer Vision, Reinforcement Learning, Edge AI, Quantization, Model optimization, Agentic systems, MCP
- **Software Development:** React, Next.js, React Native, Node.js, Express, Django, Flask, FastAPI, REST, GraphQL, gRPC, Flutter, Dart, WebSockets, Tailwind CSS, Vue, Git, GitHub, GitHub Actions, GitLab, CI/CD, APIs
- **Databases and Data Engineering:** MongoDB, PostgreSQL, Redis, MySQL, Vector databases, ETL pipelines, Pandas
- **DevOps and Cloud Infrastructure:** Docker, Kubernetes, CI CD pipelines, RabbitMQ, Linux, AWS EC2, AWS Lambda, AWS S3, GCP, Terraform, Nginx, Edge deployment, Serverless architecture, Scalable ML inference orchestration
- **Core Computer Science:** Data Structures, Algorithms, Time and Space Complexity Analysis, Object Oriented Design